Breast CARE-MA - Project Plan and Architecture Document

Final Year B.Tech IT Project Team: Partha, Atchaya, Kalaiarasi, Hariharasudhan

Project Vision

Develop a 6-agent explainable AI breast cancer diagnostic decision support system using React, FastAPI, CrewAI/n8n, PostgreSQL and open-source AI models.

System Architecture

User -> React Frontend -> FastAPI Backend -> Agent Orchestrator -> Agents 1-6 -> Database -> Report Generation -> PDF Report.

Agent Responsibilities

Agent 1 Detection, Agent 2 Classification, Agent 3 XAI/Grad-CAM, Agent 4 Evidence Retrieval, Agent 5 Risk Assessment, Agent 6 Report Generation.

Team Allocation

Partha: Architecture, Backend, Integration, Deployment. Atchaya: Detection and Classification Models. Kalaiarasi: Frontend and Explainability UI. Hariharasudhan: RAG, Risk Assessment and Report Generation.

Technology Stack

Frontend: React+Vite. Backend: FastAPI. Database: PostgreSQL/Supabase. AI: PyTorch, YOLOv8, ResNet. LLM: Llama 3 via Ollama. Workflow: n8n/CrewAI.

Development Phases

Phase 1 Requirements & Dataset. Phase 2 Model Development. Phase 3 Frontend. Phase 4 Backend APIs. Phase 5 Multi-Agent Integration. Phase 6 Testing and Deployment.

Git Workflow

main=stable, develop=integration, feature branches per member. Weekly code review and merge.

Deliverables

Research paper, source code, deployed application, PPT, report, demo video and final thesis.

16 Week Execution Plan

- Weeks 1-2: Requirements, datasets, architecture
- Weeks 3-5: Detection and classification models
- Weeks 6-8: Frontend and backend development
- Weeks 9-11: Agent implementation and integration
- Weeks 12-13: Testing and validation
- Weeks 14-15: Documentation and research paper
- Week 16: Demo and final viva preparation